

Architecture:

Architecture diagram:

Source code (written by the user)

|

Lexer/Tokeniser (transforms the source code into tokens)

|

Creation of parsing table and LL(1) parsing using a stack with AST creation during parsing

|

Semantic analysis with type checking and a linker that imports/finds imported functions from the three libraries of Snake

|

Execution of the code

- Why LL(1) parsing?

Because it is more memory efficient than for example LR(1) parsing, as well as could be more efficient for relatively small programming languages like

- Why modules for libraries?

Because they do not cause namespace confusion and won't let the user accidentally overload a core function.

- Why Haskell?

Because it is a language the structure of which really helps with creating interpreters/compilers and helps avoid unnecessary code, that would not help me learn anything new and would take up most of my time resulting in a worse end result. Additionally, I was interested in learning a functional language and this was a good project to do that.

- Why an interpreter?

Because I wanted to learn more about programming languages and how they are designed and expand my knowledge in programming.